



Documentation & Quick Start



Thank you!

Thank you for choosing this pack! We hope you create something really special with it.

*Please consider rating the package through your download list or leave a review at the store page once you're familiar with it.
Feel free to give us feedback via E-Mail info@tidalflask.com or our social media!*

Your feedback helps us focus on the right updates for the future which will be free for existing users!

*Enjoy, your **Tidal Flask** team!*





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Quick Start

Importing to Built-in RP project

After importing the Standard version into your project 2021.3.20 & above, which doesn't use any of the Scriptable render pipeline packages (URP/HDRP), it should just work™.

If you see any warnings in the Console window, try the Clear button and/or relaunch Unity. If the warnings don't disappear consult the FAQ or drop us an e-mail.

If you see any pink assets inside the Project window or in the scenes, simply select said asset -> right click -> Reimport and it should fix it. If you still encounter pink shaders, please make sure you have the correct pack version installed and that you are using a Unity version that is compatible with the pack.

Make sure you have Post Processing installed from Unity's Package Manager. If you install it after you imported the pack, reload the demoscene to get rid of possible errors.

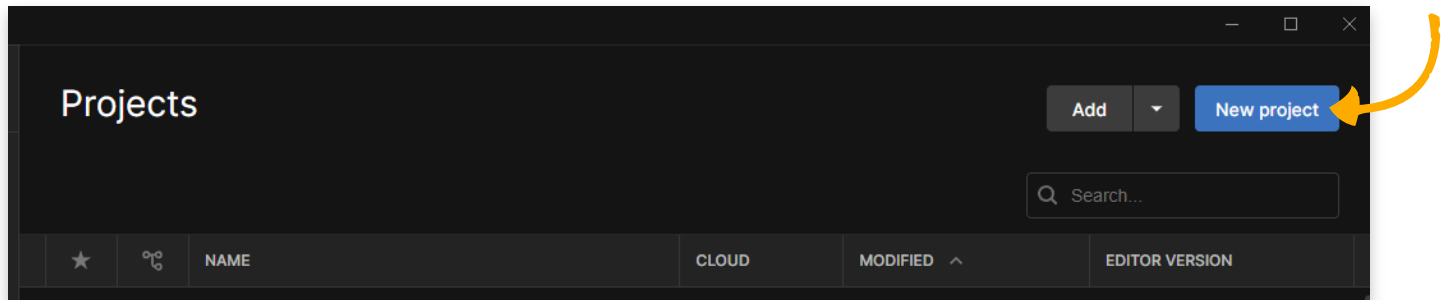
Importing to URP project

Here you will find detailed steps on how to import the package. Please note that this package only works out of the box with Unity 2021.3.20 and above.

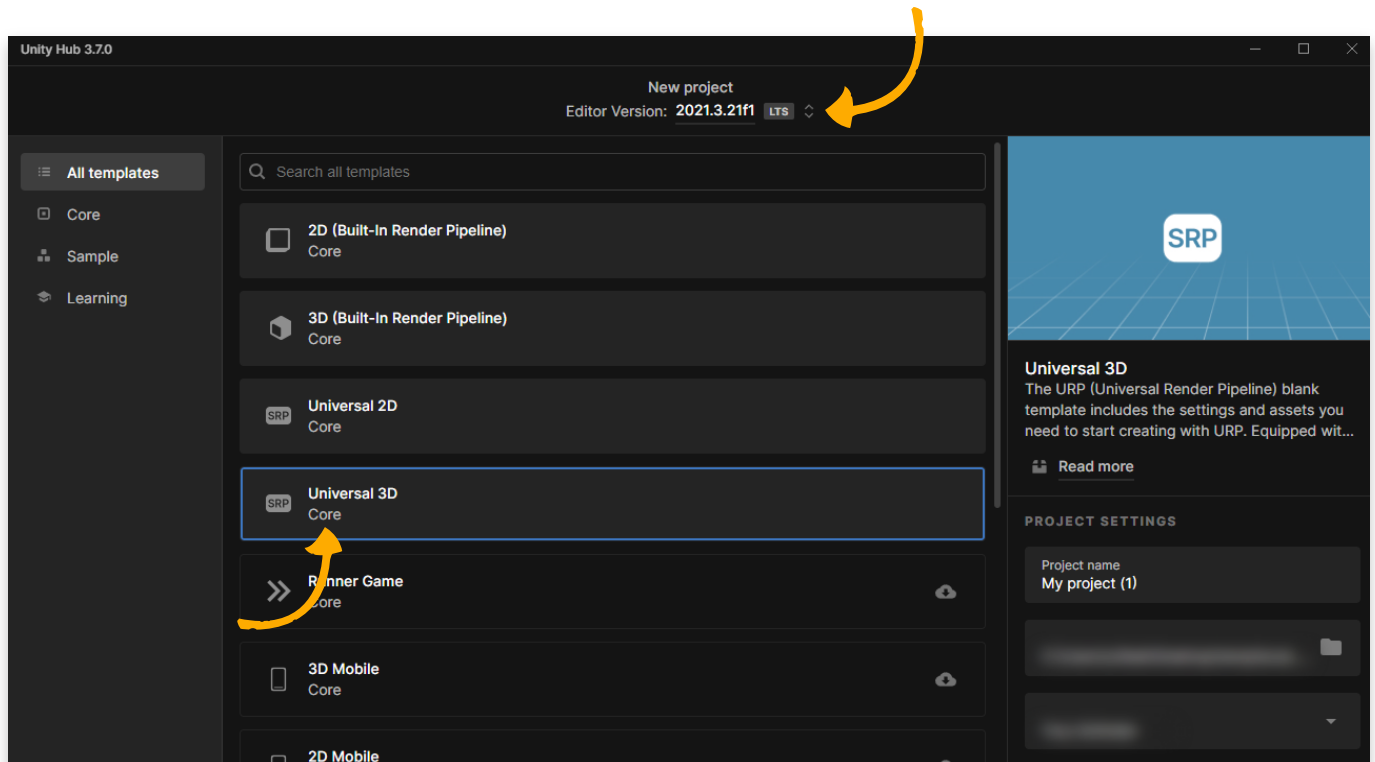


How to set up your project for URP (option 1)

We recommend to create a clean project and install the URP via the Package Manager or via Templates and import our package to this project. To do so follow the steps below:



Step 1: Click “NEW” to create a new project (for URP pick Unity 2021.3.20 or above).



Step 2: In the “Templates” select “Universal 3D”, this way everything you need for this package will be preinstalled.

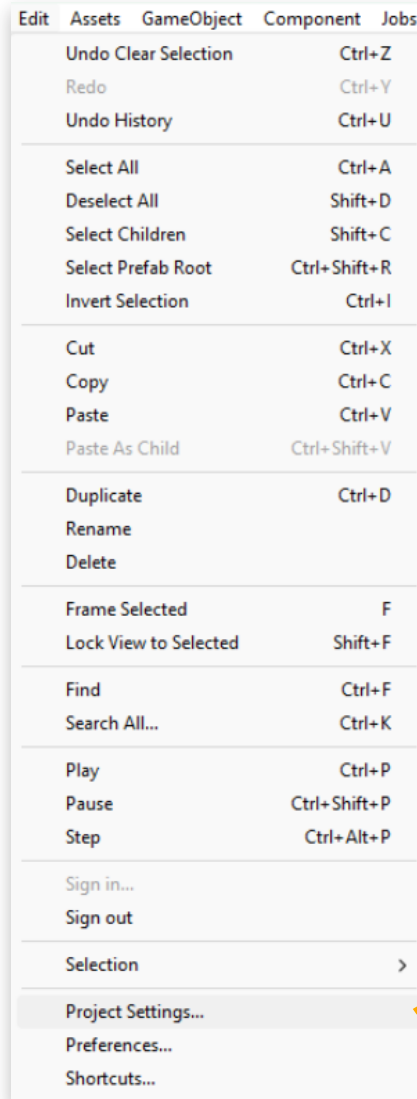


Step 3: Download the pack from the Asset Store and install the URP version. At this point you already can go to the scenes folder and select any of the scenes.

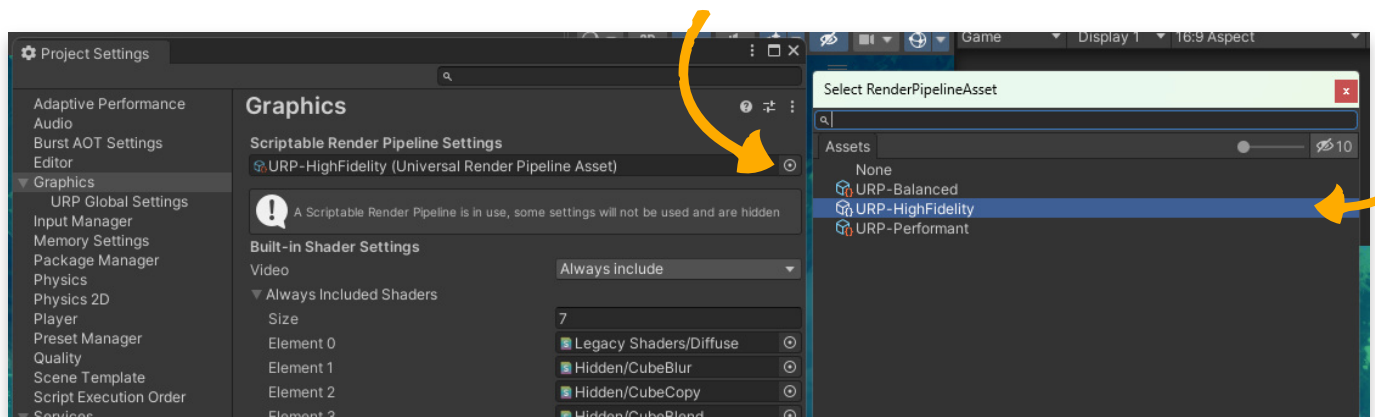
If you see any errors in the “Console”, try the “Clear” button. If the errors don’t disappear consult the FAQ or drop us an e-mail.

If you see any pink assets inside the Project window or inside the “Terrain”-object in any of the scenes, simply select the said Prefabs (inside the prefabs folder) or the Meshes (inside the 3d folder) > right click > Reimport and it should fix it.

If you still encounter pink shaders, please make sure you have the correct pack version installed, depending on the render pipeline you are using.



Step 4: After the project is loaded, go to Edit > Project Settings...

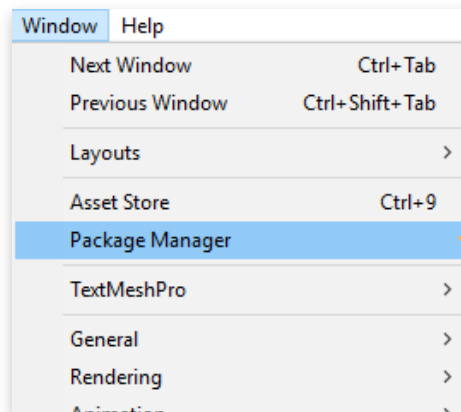


Step 5: For the Scriptable Render Pipeline Settings select “URP-HighFidelity”. These are the presets Unity preinstalled with the Template.

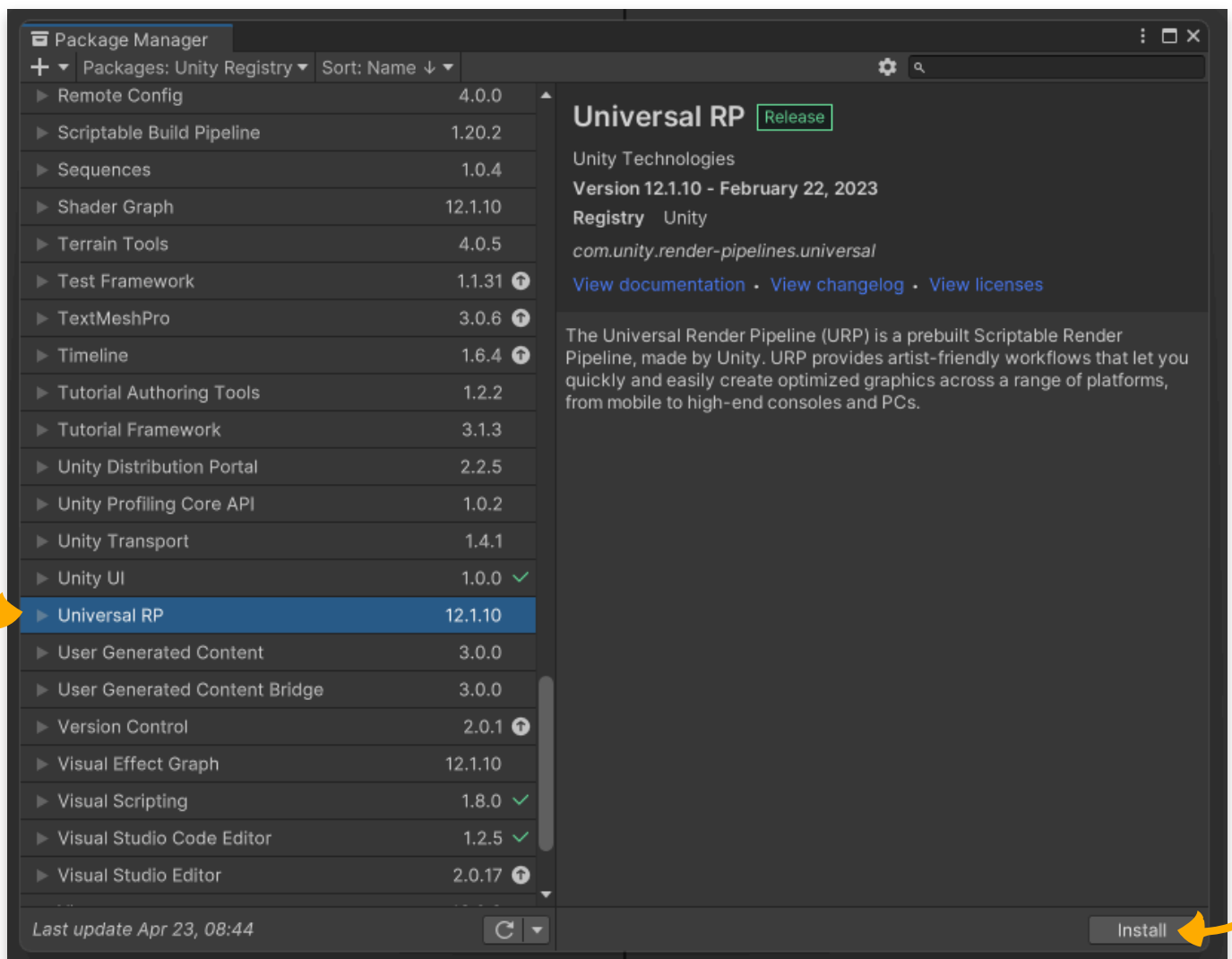


How to set up your project for URP (option 2)

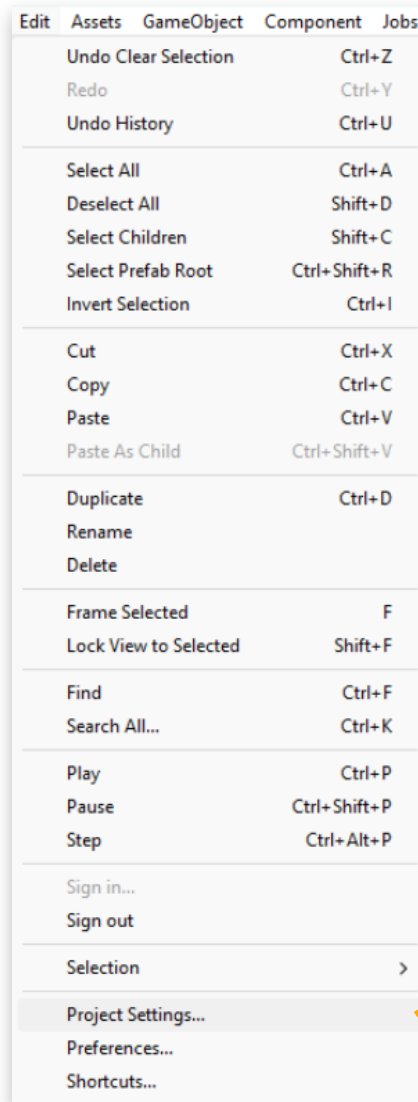
If you imported the the pack before you installed the URP please follow the steps below:



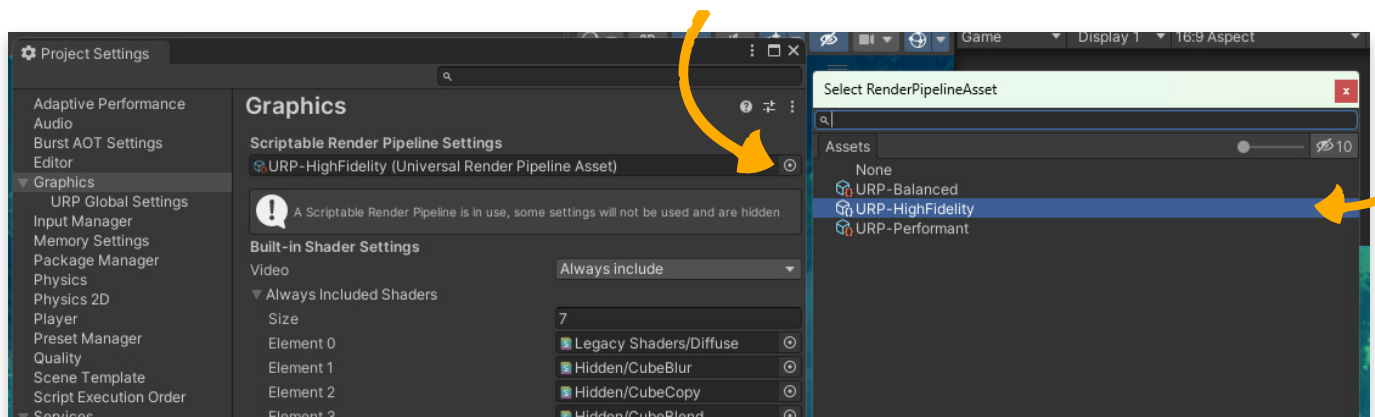
Step 1: go the Window > Package Manager.



Step 2: Select “Universal RP” asset and click “Install”.



Step 3: After the project is loaded, go to Edit > Project Settings...



Step 4: For the Scriptable Render Pipeline Settings select “URP-HighFidelity”. These are the presets Unity preinstalled with the Template.



Demo scenes and Post Processing

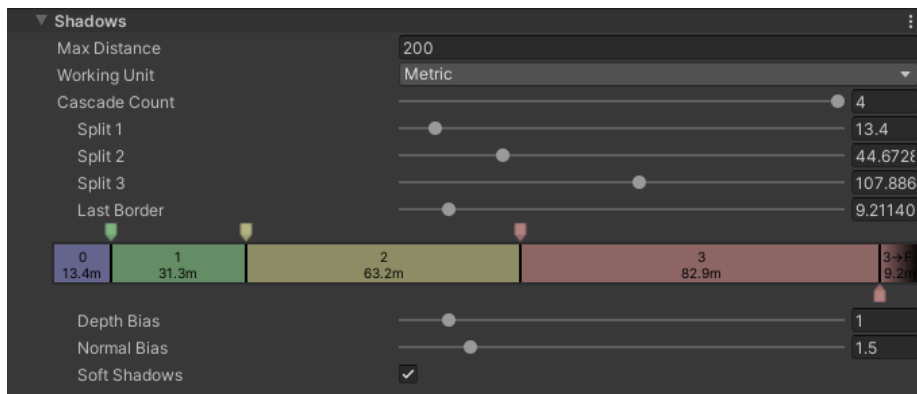
demoscene_seaside_town_level: scene with different sceneries

demoscene_seaside_town_assets: in this scene you will find all the assets within the package

All the sceneries you see in the trailer were recorded directly out of the demo scene.

Quality settings for URP

To quickly adjust any quality settings for URP please find the Render Pipeline Asset inside the /Assets/Settings folder.



Example settings for shadows

Post Processing

Inside the /Fantastic Seaside Town/Settings folder you will find a post processing file for the demo scene (Global Volume Profile). There you can adjust the post processing to your liking.

The Global Volume Profile is linked in the demo scene's Global Volume asset.



Important note: the pack was created in unity 2021.3.20, some postprocessing effects and lights may appear differently in newer versions of Unity. Adjust postprocessing (especially bloom and exposure) and the light intensity of your lights in the scene.



Assets

Meshes

Vertex maps

All foliage assets have custom vertex maps for wind movement.

LODs

Most assets have 3 LODs levels. These are already set up for you inside the corresponding prefabs. You will find all the prefabs in the /prefabs folder.

Collision

All assets have custom colliders where needed.

Lightmap UVs

All assets have a custom lightmap UV in the second channel.

Textures & Materials

You can find all the textures in the /2d folder. The materials are in the /materials folder. The _PBR textures are set up the following way:
R = Metallic, G = Roughness, B = Ambient Occlusion.

Materials with tileable textures

To adjust the look of the terrain materials, please adjust the terrain layers directly in this folder: /2d/terrain_layer

There are different classes of materials and textures in this pack depending on its use:

- Tileable terrain textures (grass, ground, stone)
- Atlases/Trims (props, foliage)
- FX (fire, smoke, water)



Shaders

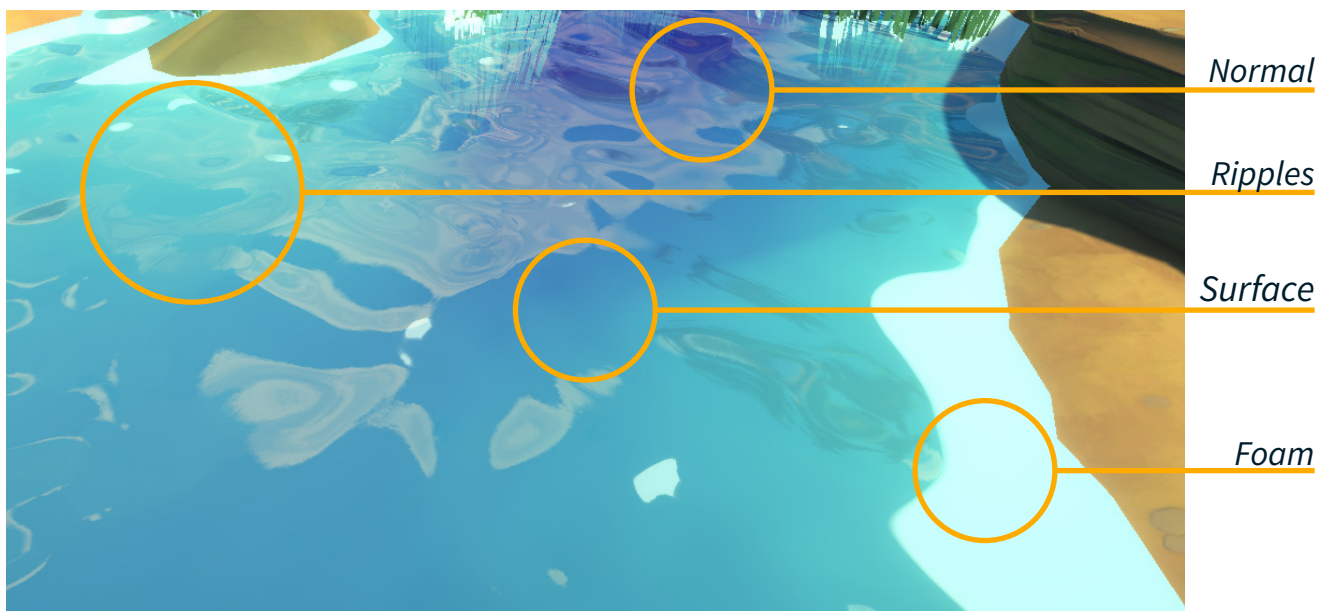
The shaders were created using “Amplify Shader Editor” and hence can **not** be opened or adjusted using “Unity’s Shader Graph”. Of course, if you have Amplify installed, you can adjust the shaders there.

The shaders are URP or Built-in, depending on which render pipeline you are using.

Water shader

The water material is defined by four main parts:

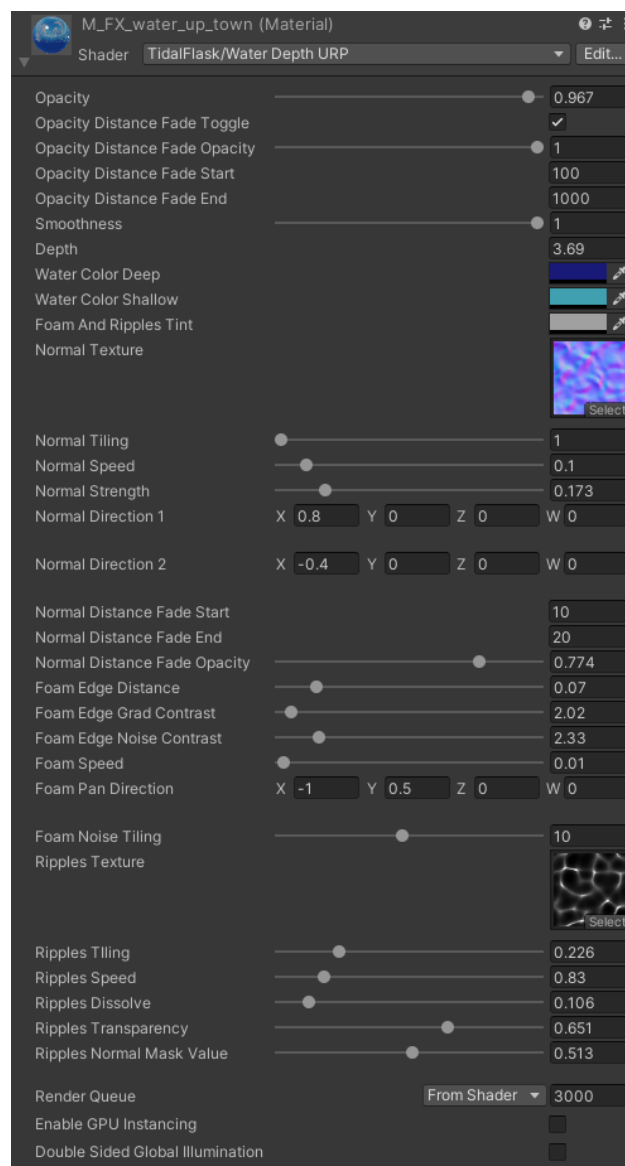
- General Surface: Defines colors, opacity and depth
- Normal: Defines the Normal of the water surface
- Foam: Creates a foam effect where meshes intersect with the water
- Ripples: Defines the ripple like highlights on the water surface



- Important note: If the water isn’t displayed correctly, make sure to enable “Depth Texture” in your Render Pipeline Asset and play around with the shadow cascades in case you are experiencing shadow issues with the water.
- If you are applying the water material to a new plane, make sure to turn off “Cast Shadows” in the inspector window of the plane.



- Opacity: Water opacity (affects ripples and foam as well)
- Opacity Distance Fade Toggle: toggle on to make the water fade out towards the distance (relative to camera position)
- Opacity Distance Fade Start/End: start/end of the fade in meters
- Smoothness: Smoothness of the water
- Depth: Depth of the water (affects Water Color Deep and Shallow)
- Water Color Deep: Deep color
- Water Color Shallow: Shallow color
- Foam And Ripples Emission: emission color, you can also tone down emission strength by bringing the color down towards darker values
- Normal Texture: normal texture
- Normal Tiling: normal texture tiling
- Normal Speed: normal offset speed
- Normal Strength: normal strength
- Normal Direction 1-2: direction for offset
- Normal Distance Fade Start/End: Defines Normal strength between a distance start and end value
- Normal Distance Fade Opacity: Opacity value to fade between texture based normal and fully neutral normal
- Foam Edge Distance: distance of the foam from shore to end of foam (visually dependant on Foam Edge Grad Contrast)
- Foam Edge Grad Contrast: contrast for foam edge
- Foam Edge Noise Contrast: contrast for noise overlay on top of foam edge gradient
- Foam Speed: speed of Foam Edge Noise
- Foam Pan Direction: Foam Edge Noise offset direction
- Foam Noise Tiling: tiling of Foam Noise
- Ripples Texture: R channel of the ripples texture
- Ripples Tiling: Tiling of the ripples texture
- Ripples Speed: offset speed of the ripples texture
- Ripples Dissolve: “contrast” of the ripples texture (dependant on the resolution/contrast of the input texture)
- Ripples Transparency: ripples transparency (affects emission)
- Ripples Noise Mask Value: value for how much the Normal texture masks the ripples (dependant on normal texture, tiling etc.)



The customisation options inside the water material.



Foliage shader

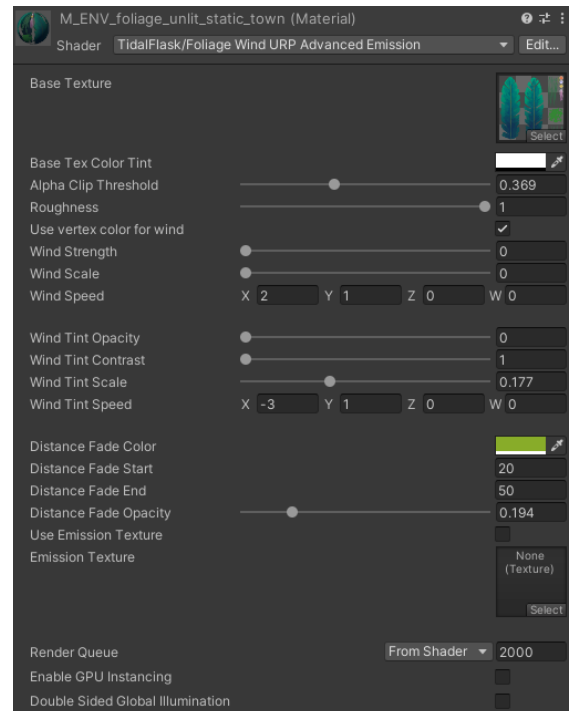
For the foliage wind movement we have included 2 shader variations in this pack:

- S_foliage_wind_URP_advanced_emission:
- S_foliage_wind_URP_advanced_lit_emission:

The advanced wind shader is defined by four main parts:

- Base Color/Texture: Defines texture and tint of the surface
- Wind Movement: Defines strength, scale and direction of the wind movement.
- Wind Tint: Creates animated highlights on the assets
- Distance Fade: Defines a distant color fade relative to camera position.

- **Difference between advanced and advanced lit:**
Lit version has the front and backfaces differently shaded depending on your lights (mostly used for fern-like assets where you would want the backface shaded. The non-lit version has both sides lit the same (mostly used for grass))
- Base Texture: Slot for the foliage texture
- Base Tex Color Tint: Defines texture tint color
- Alpha Clip Value: Defines threshold when the pixel will be opaque or transparent
- Roughness: roughness value 0-1
- Use vertex color for wind: Enable this option, if you are using a mesh with a vertex map. Fallback is UV coordinates
- Wind Strength: Strength of the deformation
- Wind Scale: Defines the density of the noise applied to the mesh
- Wind Speed: Movement direction of the noise (only edit the x and y values, z and w components are not used)
- Wind Tint Opacity: Defines transparency of the tint color
- Wind Tint Contrast: Defines contrast of the tint color
- Wind Tint Scale: Size of the noise for the tint
- Wind Tint Speed: Movement direction & speed of the noise
- Distance Fade Color: Defines the color of the fade in the distance
- Distance Fade Start: Defines start of the fade relative to the camera view
- Distance Fade End: Defines the end of the fade
- Distance Fade Opacity: Defines transparency of the distance fade
- Use Emission Texture: Toggle if you want to use the emission texture input
- Emission Texture: Slot for the emission texture



The customisation options inside the wind material.



PBR shaders

We have included 4 different PBR shaders for different usecases:

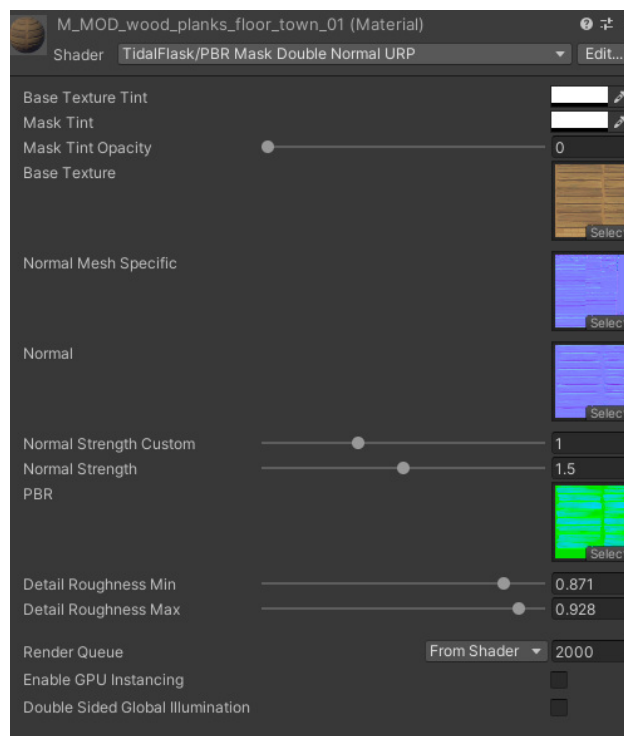
S_pbr: the default PBR shader with inputs for Base Color, Normal, and PBR map (R: metallness, G: roughness, B: ambient occlusion)

S_pbr_emission_normal: PBR shader with inputs for Base Color, Normal, PBR map (R: metallness, G: roughness, B: ambient occlusion), and Emission texture

S_pbr_mask_doublenormal: PBR shader with inputs for Base Color, Normal, additional asset specific Normal, and PBR map (R: metallness, G: roughness, B: **tint mask**)

S_pbr_mask_normal: PBR shader with inputs for Base Color, Normal, and PBR map (R: metallness, G: roughness, B: **tint mask**)

- Base Texture Tint: Tint color for base texture
- Mask Tint: Tint for color overlay (based on B channel in PBR texture)
- Mask Tint Opacity: Opacity for color overlay
- Base Texture: Slot for base texture
- Normal Mesh Specific: Mesh specific normal map
- Normal: Slot for normal map
- Normal Strength Custom: Mesh specific normal map strength
- Normal Strength: Normal map strength
- PBR: Slot for PBR texture
- Detail Roughness Min/Max: Roughness min/max, based on B channel in PBR texture
- Emission Color: Tint color for emission texture
- Use Emission Texture: Toggle to use emission texture
- Emission Texture: Slot for emission texture



PBR Mask Double Normal material



Additional emission settings for _emission_ shader



FX

Inside the /prefabs/FX folder you will find some effects to decorate your scenes. You have access to the following effects:



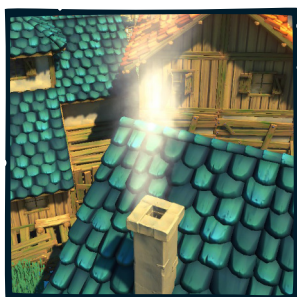
Fire



Fog



Particles Glow



Smoke

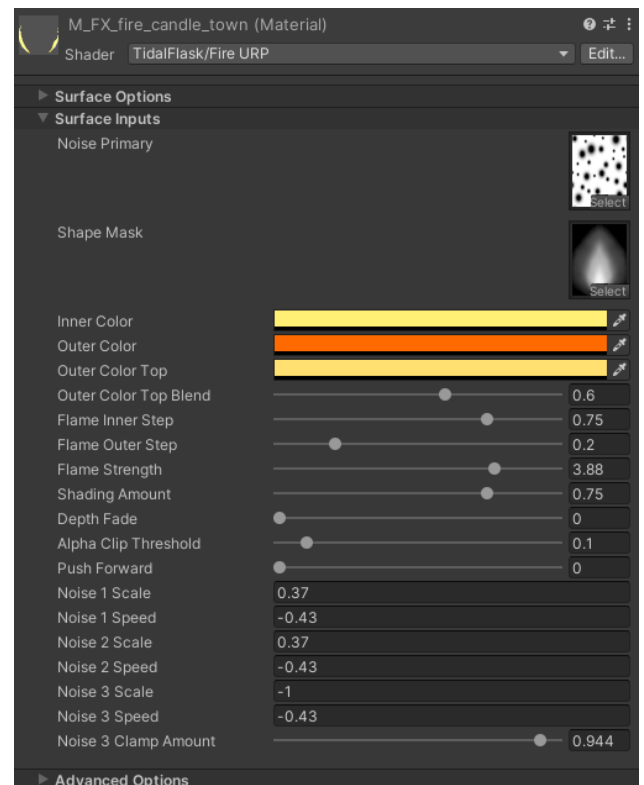


Windtrails



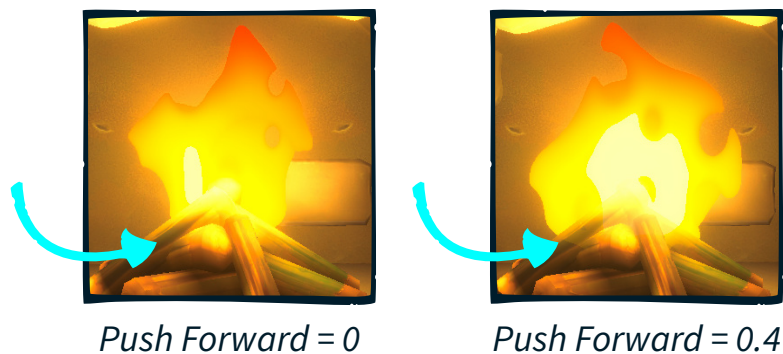
Fire FX shader

- Noise Primary: Primary noise texture
- Shape Mask: Shape mask for the flame
- Inner/Outer Color: Color for the inner/outer shape of the flame
- Outer Color Top: Color for vertical gradient of the outer color
- Outer Color Top Blend: strength of the vertical gradient
- Flame Inner/Outer Step: value selection for the inner/outer shape based on the Shape Mask
- Flame Strength: Strength control for the flame
- Shading Amount: Shading amount for the flame
- Depth Fade: Depth fade strength
- Alpha Clip Threshold: Alpha clip threshold value
- Push Forward: Billboard forward push value (this is dependant on the relative object rotation)
- Noise 1-3 Scale: Tiling value for Noise Primary
- Noise 1-3 Speed: Speed value for Noise Primary
- Noise 3 Clamp Amount: Clamp value for Noise 3



Fire FX shader - "Push Forward"

The "Push Forward" value of the fire shader is dependant on the relative object rotation. Since it's a billboarded material, even if the actual object rotation is 0, if it's nested in a parent which has a rotation $\neq 0$ this effect will have an undesired offset. To fix this rotate the "P_FX_fire_town" in the opposite direction by the same value.





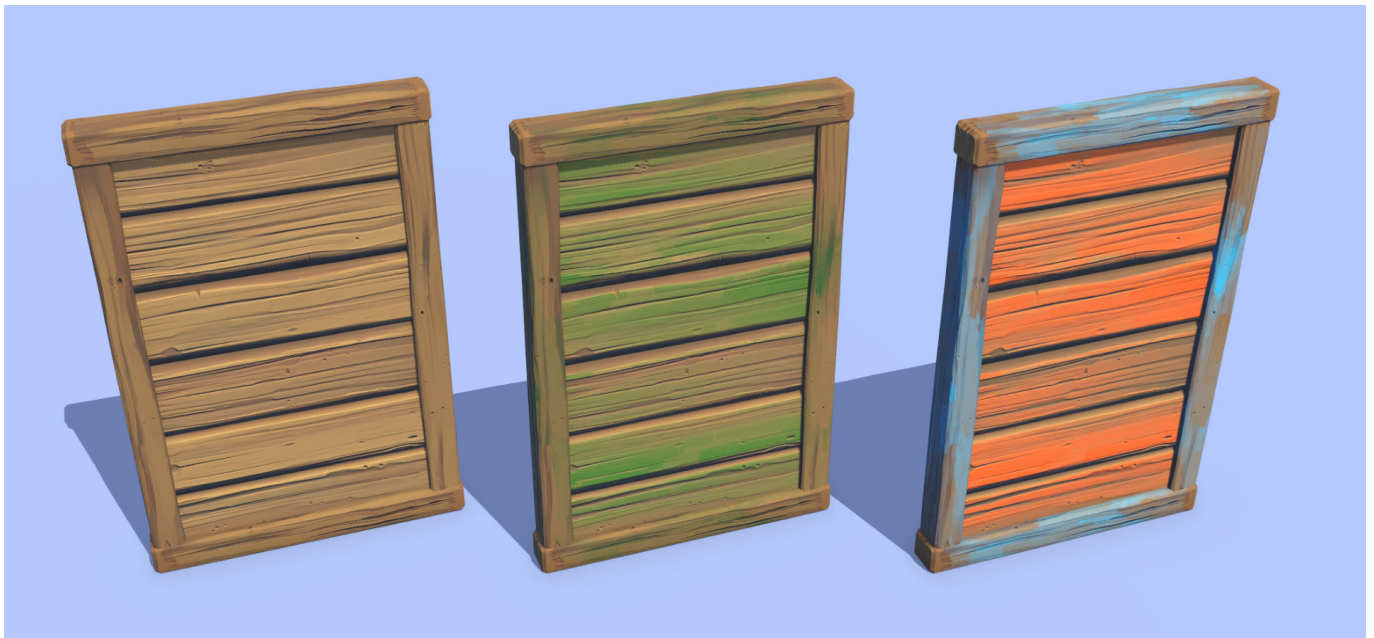
Customizing Assets

Tileable Textures

The tileable textures can be used interchangeably, be that for walls or for terrain.

Custom shaders with tint options

Every custom shader has additional tint color as an optional customization element. You want pink roofs? - You can!



Tint customization for modular assets



Scene setup

Modular assets

We have 3 levels of hierarchy for the prefabs:

level 1: Parts - individual modular elements, baseline prefabs + collision

level 2: Comps - compositions of individual elements and individual modular elements that are ment to be used as-is (e.g. Beams)

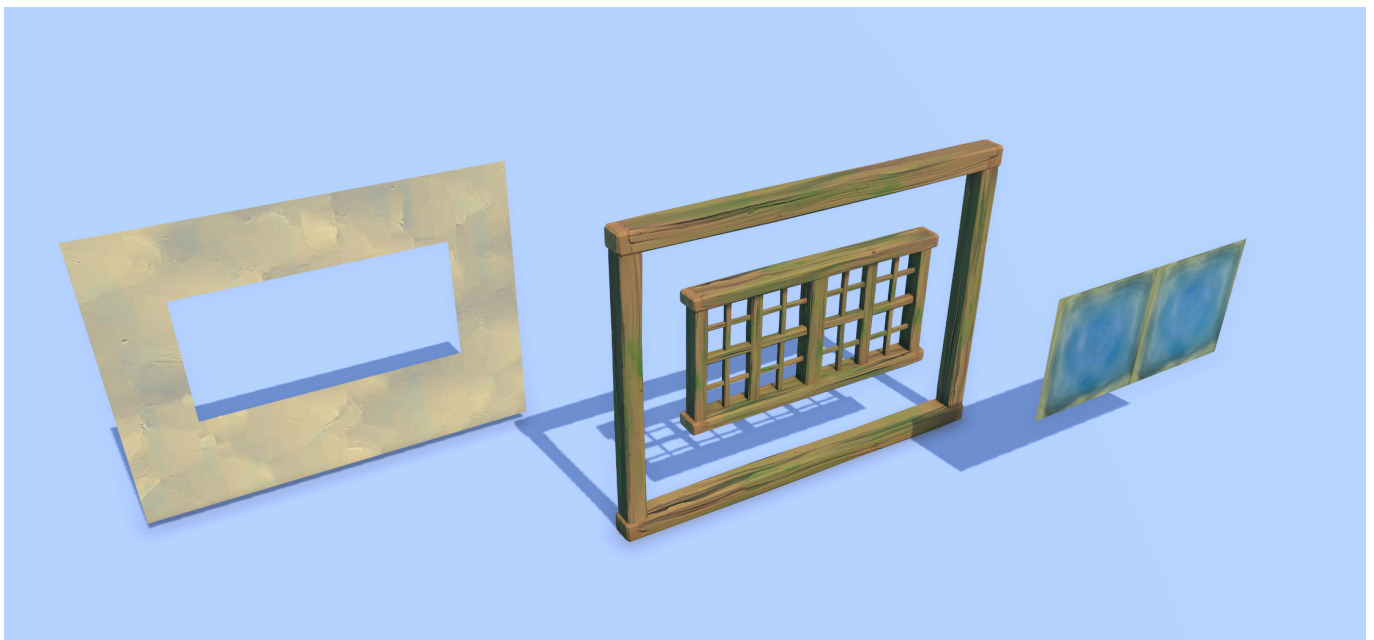
level 3: Buildings - combinations of Comps and Parts

Adjusting anything on a lower level of the hierarchy will automatically propagate up (except there are already changes on a higher level). Adjusting anything in a higher level will not automatically propagate down and you will have to manually apply your changes if you want them to be in the lower level prefab.

Read more into the Nested Prefab workflow here (select Unity version on the top left):

<https://docs.unity3d.com/6000.1/Documentation/Manual/NestedPrefabs.html>

Assets/Fantastic Seaside Town/Prefabs/Modular_parts



Example: Individual pieces of a wall (Wall, Frame, Glass)



Assets/Fantastic Seaside Town/Prefabs/Modular_comps



Example: Comp of a wall asset

Assets/Fantastic Seaside Town/Prefabs/Buildings



Example: Complete building



Working with modular assets

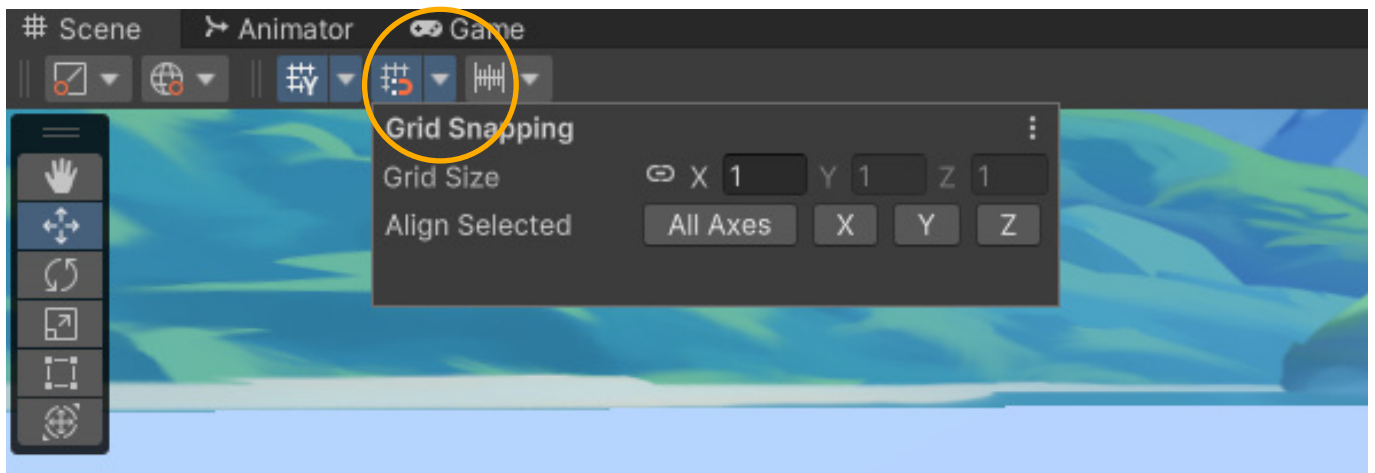
Snapping

The modular elements are following a 4x4 grid. The walls are 3 units high.

Generally speaking every modular asset needs to snap on nondecimal numbers. You will notice, that for this rule there are some exceptions when combining certain elements that do not naturally match.

You can snap assets by holding Control (Command) key while moving or rotating objects.

Additionally, in your Scene Tab, on the top right you can expand the snapping window. Press “All Axes” button when placing any of the modular elements into the scene to perfectly align them on the grid.



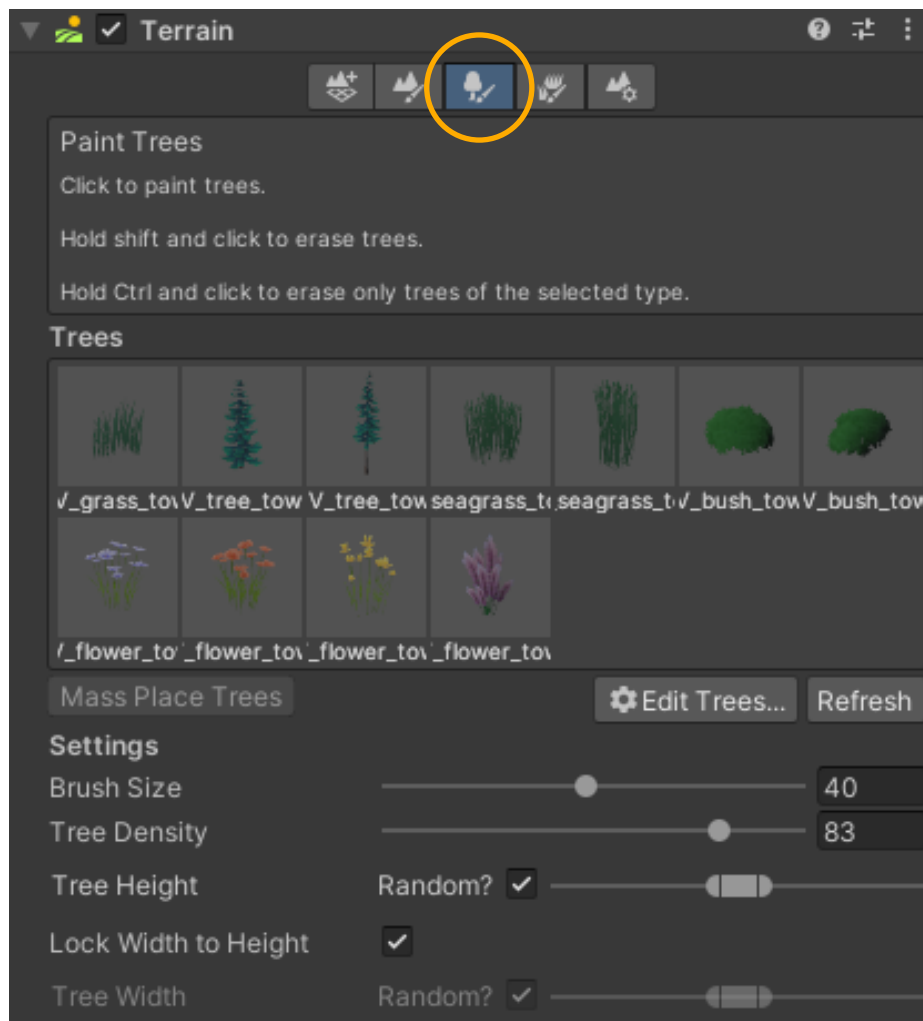
Grid Snapping menu



Environment setup - Terrain Tool

All assets are compatible with Unity's Terrain Tool.

Select your Terrain and in the Inspector tab, simply add the prefab as a “Tree” in the section “Paint Trees”. You then can paint the selected prefab on your Terrain.



With the “Paint Trees” option you can add various foliage assets and paint them on your terrain.





Support

FAQ

Will there be updates to the package?

Yes. We plan to update all our packages as soon as there is a relevant update or if the community asks for adjustments.

Can you give support to users if something doesn't work?

Yes, but first please read through this document and if you still need help with something related to this package, feel free to contact us.

A list of errors shows up in a shader.

Try reimporting the shader (in project tab > right-click on the shader > Reimport). We are aware of some shader warnings showing up, which don't seem to actually break the shader. So simply clearing the warning in the console tab should fix the problem.

I opened the project for the first time and everything is pink. When I select a material, the shader says "Hidden/InternalErrorShader"

This is the case when your project doesn't use the same render pipeline as the pack version you installed. Starting on page 4 you will find all the steps needed to properly set up your project.

I opened the project for the first time and in the Console I get the error "A tree couldn't be loaded because the prefab is missing"

This is a known Unity bug (importing a package that has terrain and trees in it) and has nothing to do with the package. Simply press "Clear" in the "Console" tab and it won't appear again.





I imported the package but some assets still appear pink in the scene...

Make sure you installed the correct render pipeline version of our pack. After opening a scene it's still possible, that some assets are pink. If that is the case, do the following:

- In the Hierarchy window select "Terrain"
- In the "Paint Details" tab double click on any asset
- Click on the circle next to the asset which was added in the "Detail" panel
- Re-add the same asset and the scene should look normal again

I imported the package but some assets still appear pink in the Project window...

If you see any pink assets inside the Project window or inside the "Terrain"-object in any of the scenes simply select the said Prefabs (inside the prefabs folder) or the Meshes (inside the 3d folder) > right click > Reimport and it should fix it.

I'm using Unity version older than 2021.3.20 and the scene assets have shadow errors and/or pink materials and/or the terrain isn't showing.

The new URP shaders are created in Unity 2021.3.20 and are not backwards compatible. The errors is created by the shadow cascades settings in the render pipeline asset. You can set the Cascades option in your render pipeline asset to "No Cascades".





Contact & Support

Visit our page for updates and more packages in the future:

<https://tidalflask.com/>

Contact us if you didn't find an answer to your questions:

info@tidalflask.com

Social Media



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<https://x.com/TidalFlask>



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